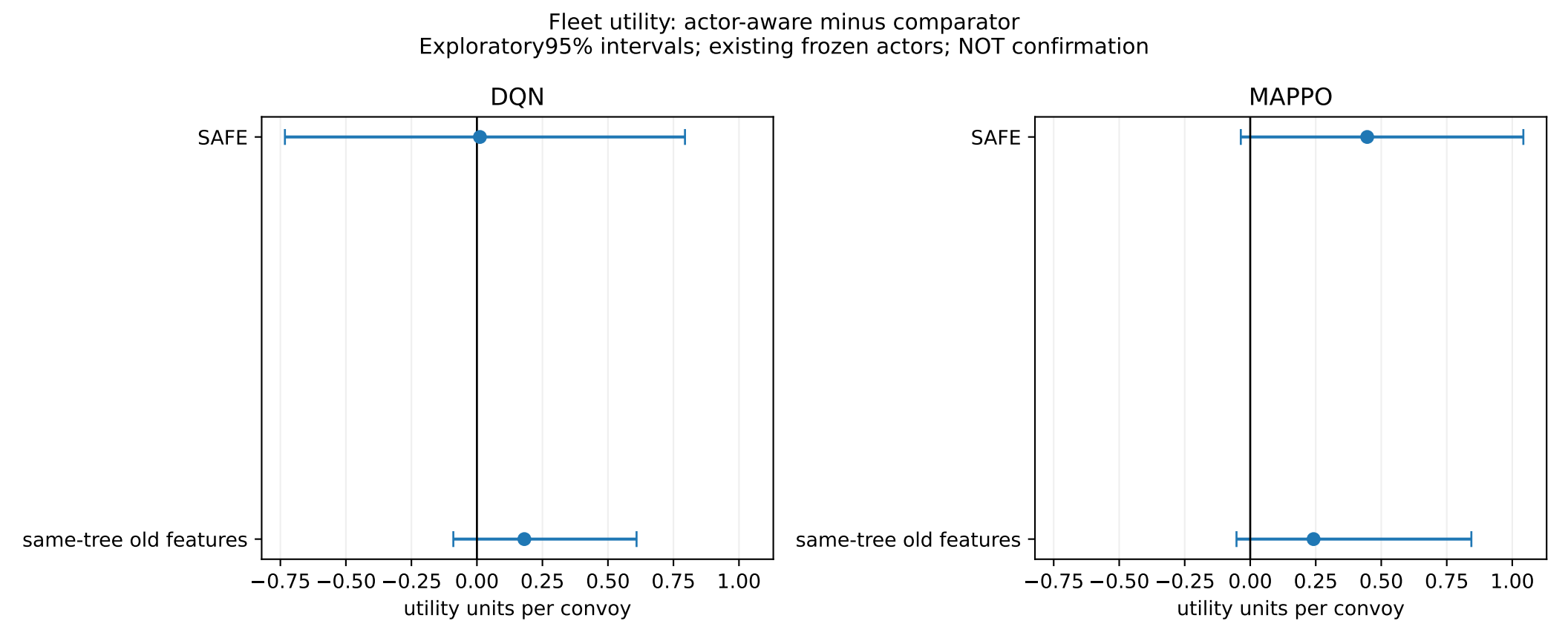
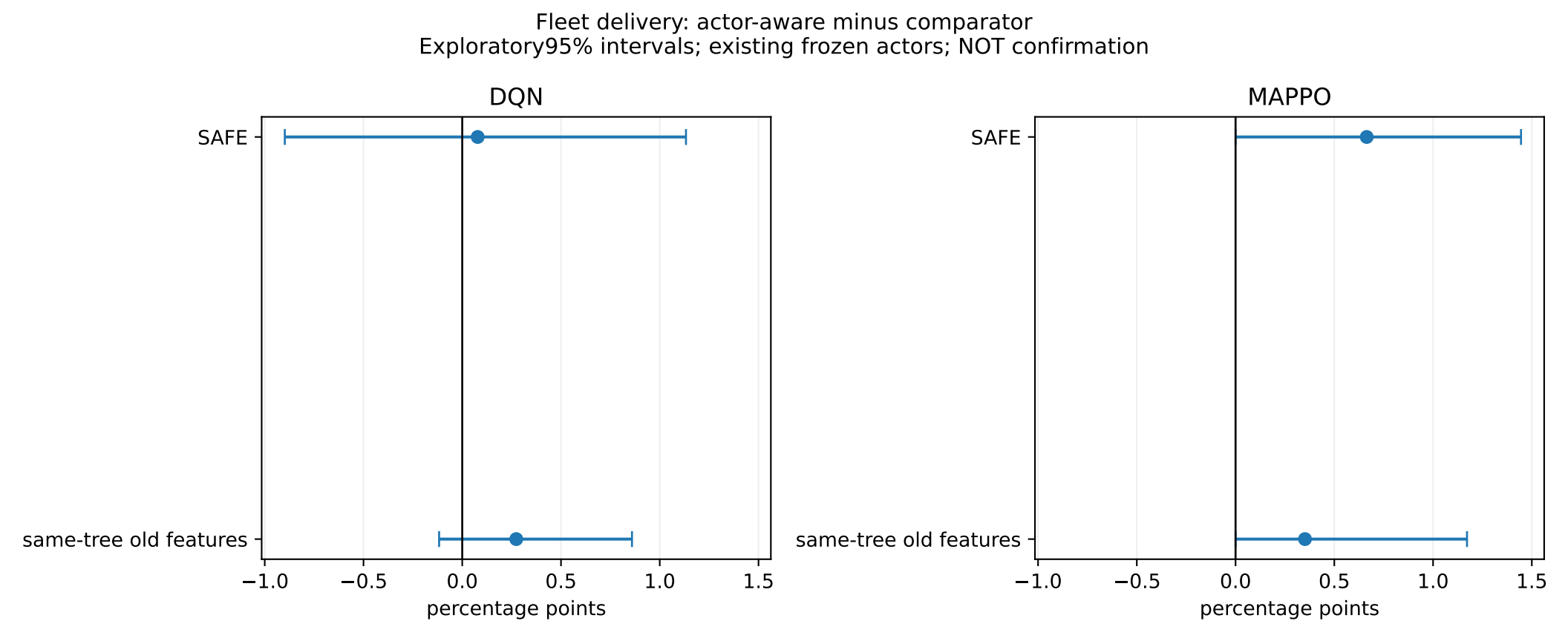
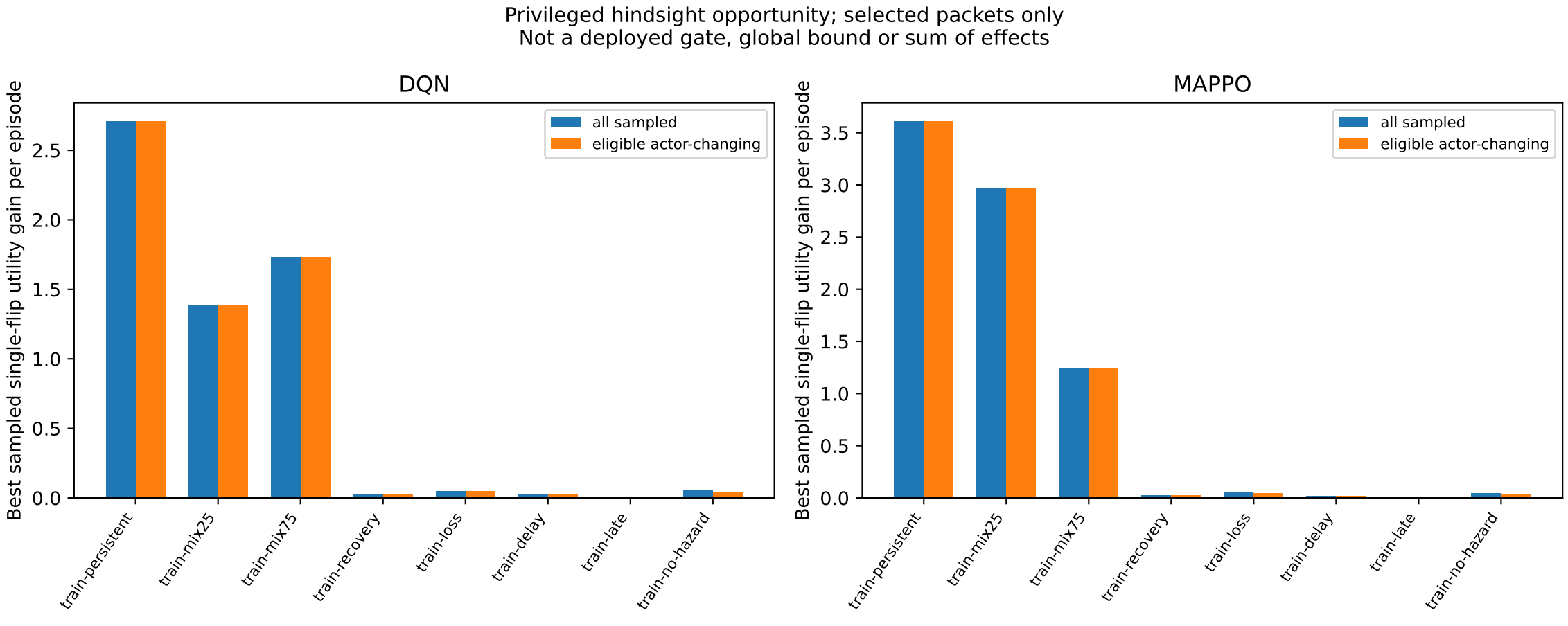
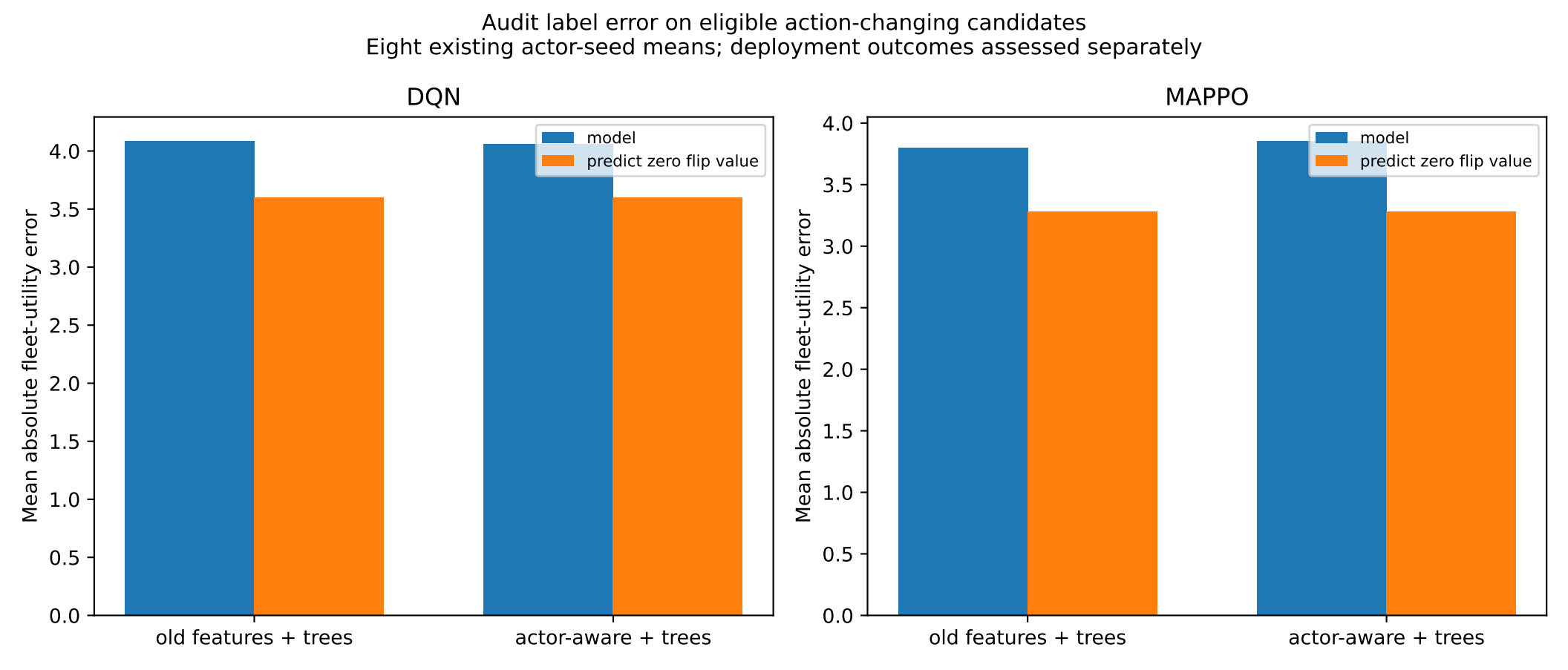
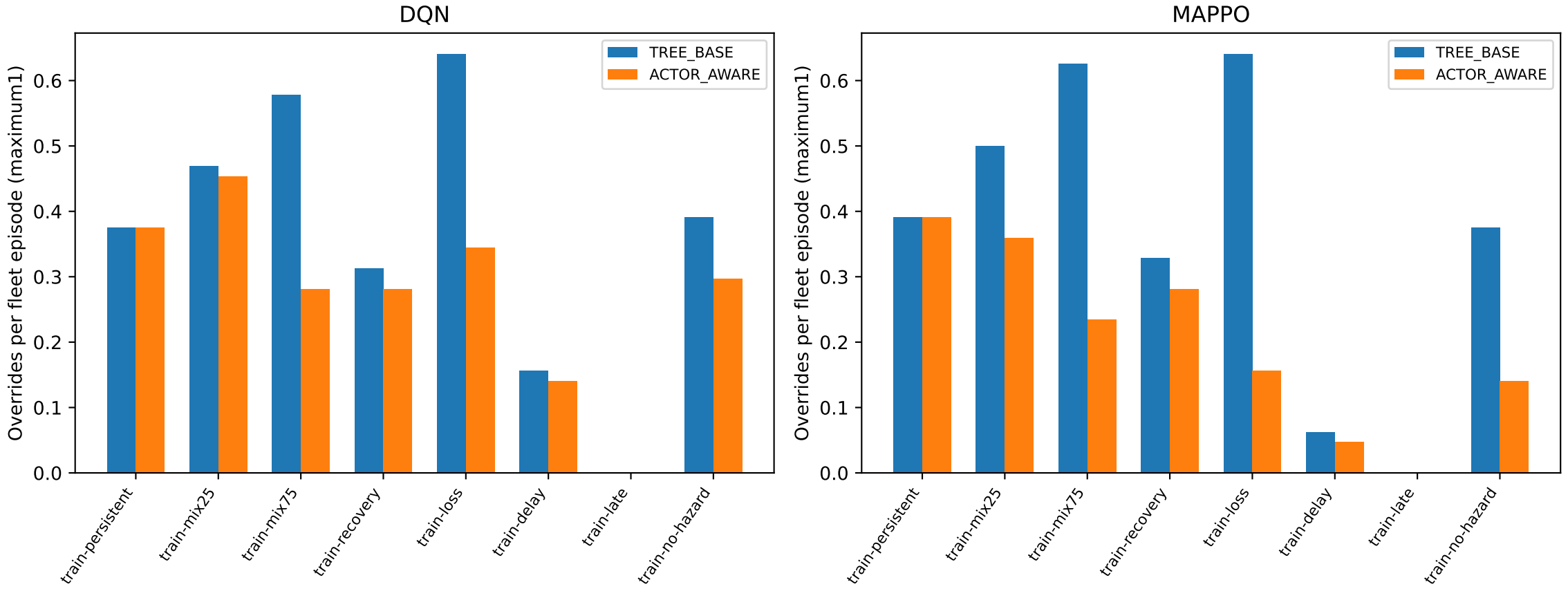
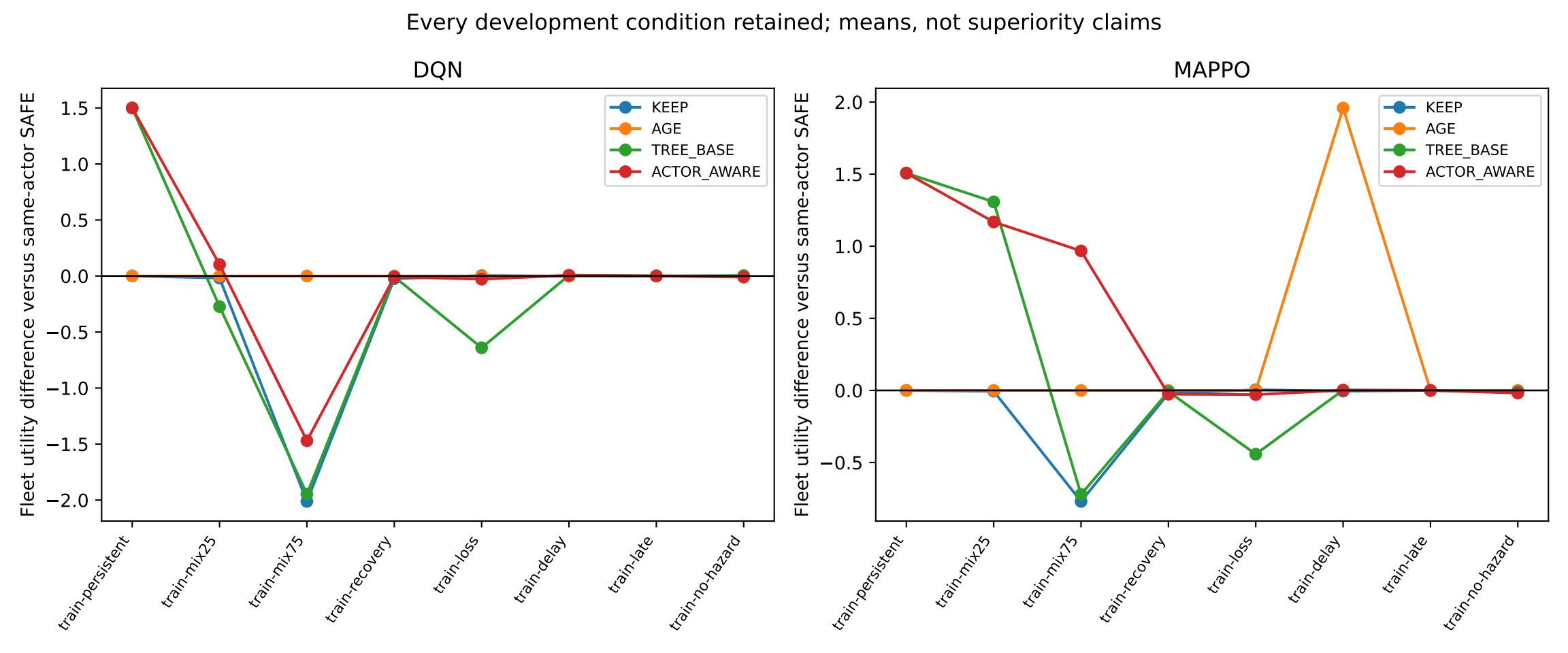

Fleet utility: actor-aware minus comparator
Exploratory95% intervals; existing frozen actors; NOT confirmation
DQN
MAPPO
SAFE
same-tree old features
−0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
utility units per convoy


Fleet delivery: actor-aware minus comparator
Exploratory95% intervals; existing frozen actors; NOT confirmation
DQN
MAPPO
SAFE
same-tree old features
−1.0
−0.5
0.0
0.5
1.0
1.5
percentage points


Privileged hindsight opportunity; selected packets only
Not a deployed gate, global bound or sum of effects
DQN
MAPPO
Best sampled single-flip utility gain per episode
all sampled
eligible actor-changing
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
train-persistent
train-mix25
train-mix75
train-recovery
train-loss
train-delay
train-late
train-no-hazard


Audit label error on eligible action-changing candidates
Eight existing actor-seed means; deployment outcomes assessed separately
DQN
MAPPO
model
predict zero flip value
Mean absolute fleet-utility error
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
old features + trees
actor-aware + trees


Actual deployment authority used, including abstention
DQN
MAPPO
TREE_BASE
ACTOR_AWARE
Overrides per fleet episode (maximum1)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
train-persistent
train-mix25
train-mix75
train-recovery
train-loss
train-delay
train-late
train-no-hazard


Every development condition retained; means, not superiority claims
DQN
MAPPO
Fleet utility difference versus same-actor SAFE
KEEP
AGE
TREE_BASE
ACTOR_AWARE
train-persistent
train-mix25
train-mix75
train-recovery
train-loss
train-delay
train-late
train-no-hazard